PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Adebusola Henry
Program/Batch	B084-Networkwalks
Date	13 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

---

1. LIABILITY DISCLAIMER

The activities documented in this report were carried out for educational and practical learning purposes as part of the NetworkWalks Cybersecurity Program, Batch B084.

I conducted the footprinting exercise on `networkwalks.com` with written permission. I performed the network scanning exercise on my own local network.

All activities were limited to the authorised targets and environments. The information presented in this report reflects the procedures and observations from the practical exercises and does not represent a full penetration test or confirmation of security vulnerabilities.

2. INTRODUCTION

As part of Week 2 of the NetworkWalks Cybersecurity Program, I participated in practical exercises focused on footprinting and network scanning. These exercises introduced me to techniques used in information gathering and host discovery during cybersecurity assessments.

The practical work covered two modules: PM1, which focused on footprinting and reconnaissance, and PM5, which focused on identifying active hosts within a local network.

During PM1, I used different Kali Linux tools to gather information about an authorised domain. These tools helped me understand how domain registration details, DNS information, web technologies, HTTP responses, and web application firewall configurations can be examined.

For PM5, I worked on Windows network configuration and Zenmap to identify my local network subnet and discover active devices that are connected to the network.

This report documents the activities I performed, the tools and commands I used, the observations made, and the knowledge gained from both practical exercises.

3. TOOLS AND ENVIRONMENT USED

The following tools and environments were used during the practical exercises:

Tool / Environment	Purpose
Kali Linux	Operating system used for footprinting activities
WHOIS	Retrieve domain registration information
WhatWeb	Identify technologies used by a website
nslookup	Query DNS information
curl	Examine HTTP responses from a web server
wafw00f	Check for web application firewall presence
dnsrecon	Perform DNS reconnaissance
Windows Command Prompt	Examine local network configuration
ipconfig	Identify local IP address and subnet information
Zenmap	Perform graphical network scanning and host discovery
Nmap	Network scanning engine used by Zenmap

4. ACTIVITIES PERFORMED

4.1 PM1 – FOOTPRINTING AND RECONNAISSANCE

Objective

The objective of this exercise was to understand how publicly accessible information about a domain can be gathered using different reconnaissance tools.

The target domain for the exercise was `networkwalks.com`, and the activity was performed with written permission.

4.1.1 WHOIS Lookup

I started the exercise by using WHOIS to retrieve registration-related information about the target domain.

Command used:

```
None
whois networkwalks.com
```

This command returned domain registration information that could help identify details such as the registrar, registration dates, domain status, and name servers where available.

Experience and learning:

This activity introduced me to the importance of domain registration information during the reconnaissance phase. I learned that WHOIS can provide useful background information about a target before proceeding with further technical analysis.

4.1.2 WhatWeb Scan

I used WhatWeb to identify the technologies and services associated with the target website.

Command used:

```
None
whatweb networkwalks.com
```

WhatWeb helped examine the website's publicly observable technology information, which may include web server details, frameworks, content management systems, and other technologies where detectable.

Experience and learning:

Through this exercise, I was able to understand how technology identification can contribute to reconnaissance. It also showed me how security professionals can use automated tools to gather information about a website's technology stack.

4.1.3 DNS Lookup Using nslookup

I used nslookup to query DNS information associated with the target domain.

Command used:

```
None
nslookup networkwalks.com
```

The command allowed me to examine DNS resolution and observe the information that was returned by the configured DNS resolver.

Experience and learning:

I learned how DNS translates domain names into IP addresses and why DNS information is important during reconnaissance. I also gained practical experience using command-line tools to investigate domain resolution.

4.1.4 HTTP Response Analysis Using curl

I used curl to send a request to the target website and examine its HTTP response.

Command used:

```
None
curl -i networkwalks.com
```

The `-i` option displays the response headers alongside the response content.

Experience and learning:

This exercise helped me understand that communication between a browser and a web server involves HTTP requests and responses. I also learned that response headers can provide useful information about how a web server handles requests.

4.1.5 Web Application Firewall Detection Using wafw00f

I used wafw00f to check whether the target website was protected by a Web Application Firewall (WAF).

Command used:

```
None
wafw00f networkwalks.com
```

The tool examined the target's responses to determine whether a WAF could be detected.

Experience and learning:

I learned about the role of Web Application Firewalls in monitoring and filtering potentially harmful web traffic. The exercise also introduced me to a tool used to investigate WAF presence during reconnaissance.

4.1.6 DNS Reconnaissance Using dnsrecon

I used dnsrecon to perform further DNS reconnaissance on the target domain.

Command used:

```
None
dnsrecon -d networkwalks.com
```

The tool performed DNS information gathering and examined records and configurations that could be identified through its queries.

Experience and learning:

This activity helped me understand that DNS reconnaissance involves more than simply resolving a domain name. Different DNS queries can provide additional information about a domain's infrastructure and configuration.

PM1 Summary

The PM1 exercise gave me practical experience using six different Kali Linux tools for domain footprinting and reconnaissance.

I gained an understanding of how WHOIS, WhatWeb, nslookup, curl, wafw00f, and dnsrecon serve different purposes during information gathering. I also learned that information collected from multiple tools can provide a broader picture of a target's publicly observable infrastructure.

4.2 PM5 – NETWORK SCANNING AND HOST DISCOVERY WITH ZENMAP

Objective

The objective of this exercise was to identify active devices within my local network using Zenmap and understand how network scanning can be used for host discovery.

4.2.1 Identifying My Local IP Address

I started by opening Windows Command Prompt and running the following command:

```
None
ipconfig
```

This command displayed the network configuration of my computer.

From the output, I identified the following details:

Network Information	Value
Local IPv4 Address	10.89.72.212
Subnet Mask	255.255.255.0
Network Subnet	10.89.72.0/24

I used the IP address and subnet mask to determine the network range that would be scanned.

4.2.2 Understanding the Network Subnet

The subnet mask `255.255.255.0` corresponds to `/24` in CIDR notation.

Using my local IP address of `10.89.72.212`, I identified the network subnet as:

```
None
10.89.72.0/24
```

This represents the local network range used for the host discovery exercise.

4.2.3 Performing Host Discovery with Zenmap

After identifying the subnet, I opened Zenmap and configured a host discovery scan for the local network.

The Nmap command used was:

```
None
nmap -sn 10.89.72.0/24
```

The `-sn` option performs host discovery without carrying out a port scan.

The scan was used to identify devices that responded as active hosts within the specified network range.

4.2.4 Results and Observations

The scan identified two active hosts on the local network.

Host IP Address	MAC Address
10.89.72.202	36:52:8D:02:D1:6F
10.89.72.212	Not available

One of the discovered hosts displayed a MAC address, while the other did not have a MAC address available in the scan results.

4.2.5 Experience and Learning

This exercise helped me understand how devices communicate within a local network and how host discovery can be used to identify active devices.

I also learned how to use the `ipconfig` command to identify my computer's network configuration, interpret an IPv4 address and subnet mask, and determine the corresponding network range.

Using Zenmap gave me practical experience with a graphical interface for Nmap and helped me understand the purpose of the `-sn` option during network discovery.

5. RISK ANALYSIS / POTENTIAL IMPACT

The practical exercises demonstrated how information gathered during reconnaissance and network scanning can be useful in understanding an organisation's exposed infrastructure.

The following are potential security implications of the information-gathering techniques explored:

Domain Information Exposure: Publicly available registration and DNS information may provide useful background details about an organisation's internet infrastructure.

Technology Identification: Identifying web technologies can help security professionals understand the technologies in use and assess whether they require further security review.

HTTP Information Exposure: HTTP response headers may reveal information about server behaviour and configuration. Depending on the information disclosed, this may help an attacker understand the web environment.

Network Visibility: Host discovery can reveal active devices within a network. If an unauthorised person gains access to a network, such information could help them identify potential targets for further investigation.

These are potential risks associated with the techniques studied. The exercises did not establish that the target domain or my local network contained exploitable vulnerabilities.

6. RECOMMENDATIONS

Based on the knowledge gained from the practical exercises, the following security recommendations are relevant:

1. Organisations should regularly review their domain registration and DNS configurations.
2. Web technologies and server software should be kept updated to reduce exposure to known vulnerabilities.
3. Web servers should be configured to avoid unnecessarily disclosing sensitive information through HTTP response headers.
4. Appropriate Web Application Firewall protection should be considered for publicly accessible web applications.
5. Organisations should maintain an inventory of devices connected to their networks.
6. Network access should be restricted to authorised users and devices.
7. Regular network assessments should be conducted to identify unexpected devices and possible security weaknesses.

7. CONCLUSION

The Week 2 practical exercises provided an opportunity to apply some of the fundamental techniques used in cybersecurity reconnaissance and network discovery.

Through PM1, I gained practical experience using six Kali Linux tools to examine domain registration information, website technologies, DNS information, HTTP responses, and possible Web Application Firewall protection.

Through PM5, I learned how to identify my local IP address and subnet using Windows Command Prompt and perform host discovery with Zenmap. The scan identified two active hosts within the local network range.

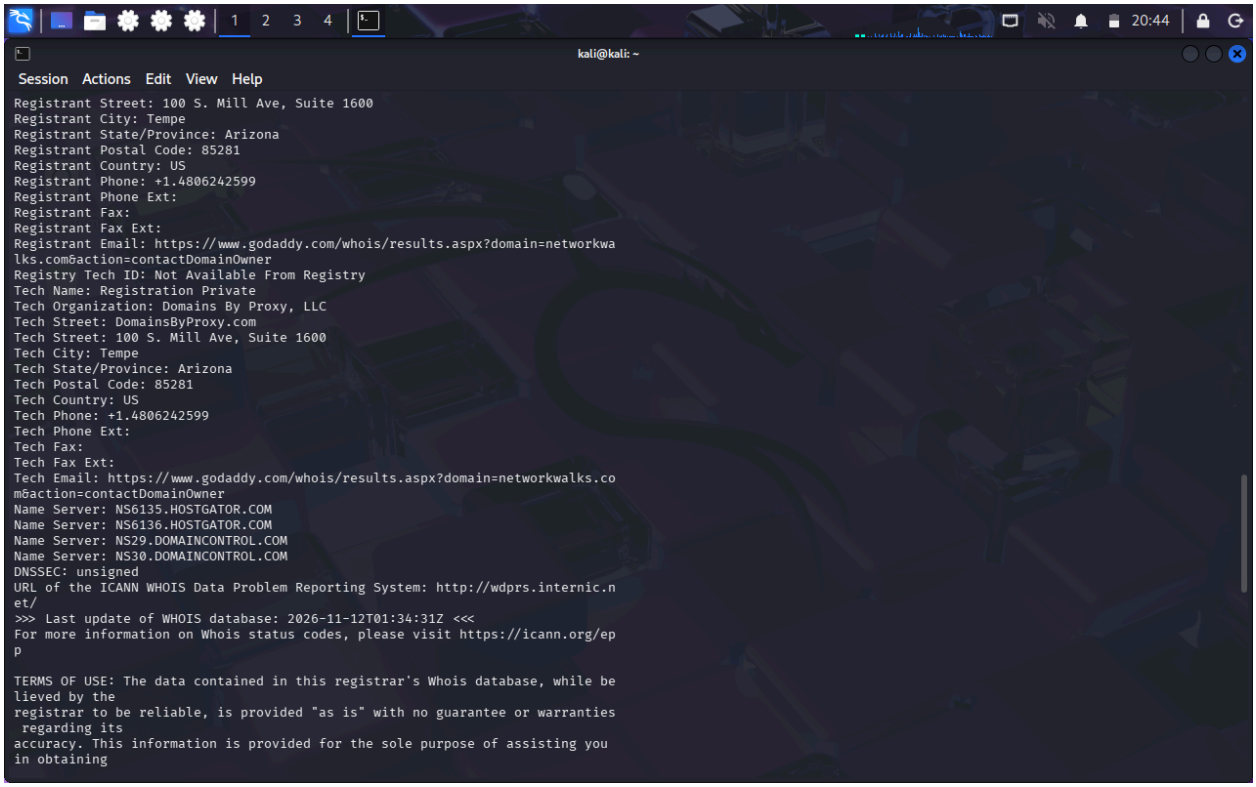
Overall, the exercises improved my understanding of information gathering, DNS, HTTP communication, IP addressing, subnetting, and network host discovery. They also helped me understand how these techniques contribute to the early stages of a cybersecurity assessment.

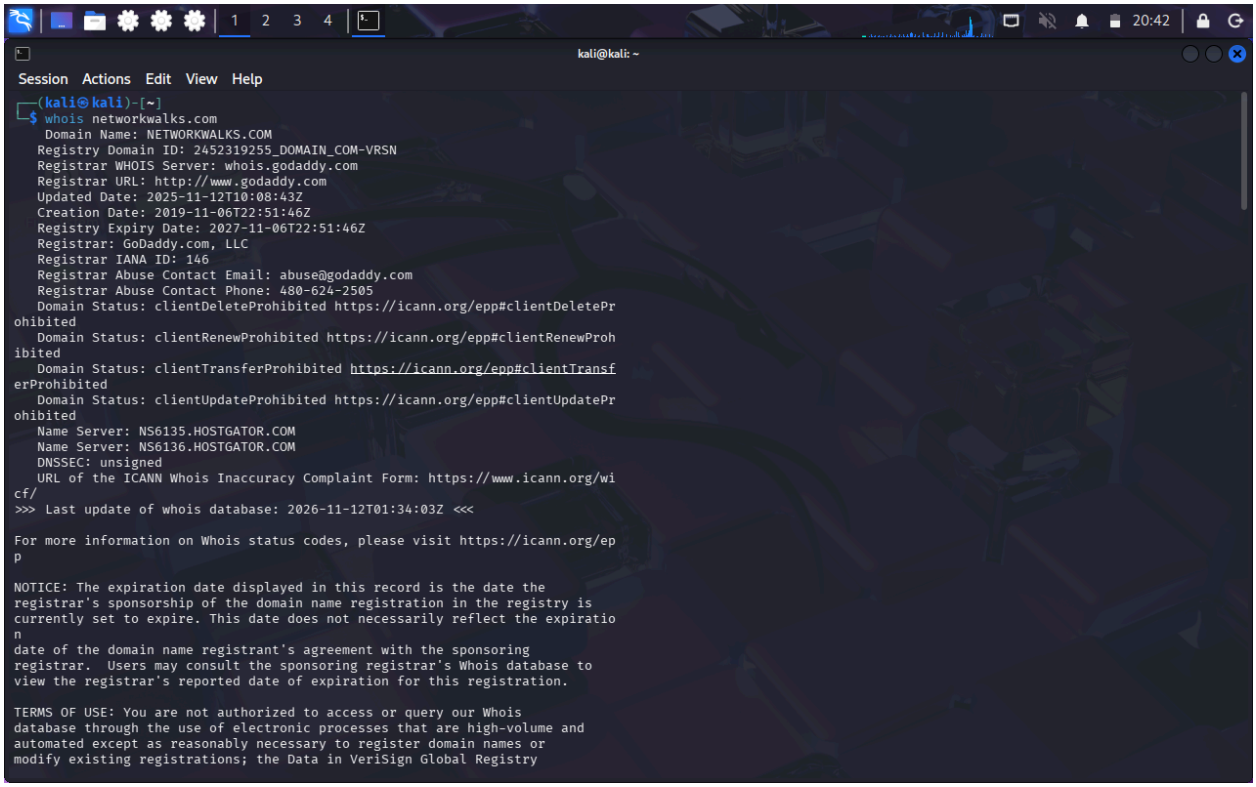
8. EVIDENCE COLLECTED

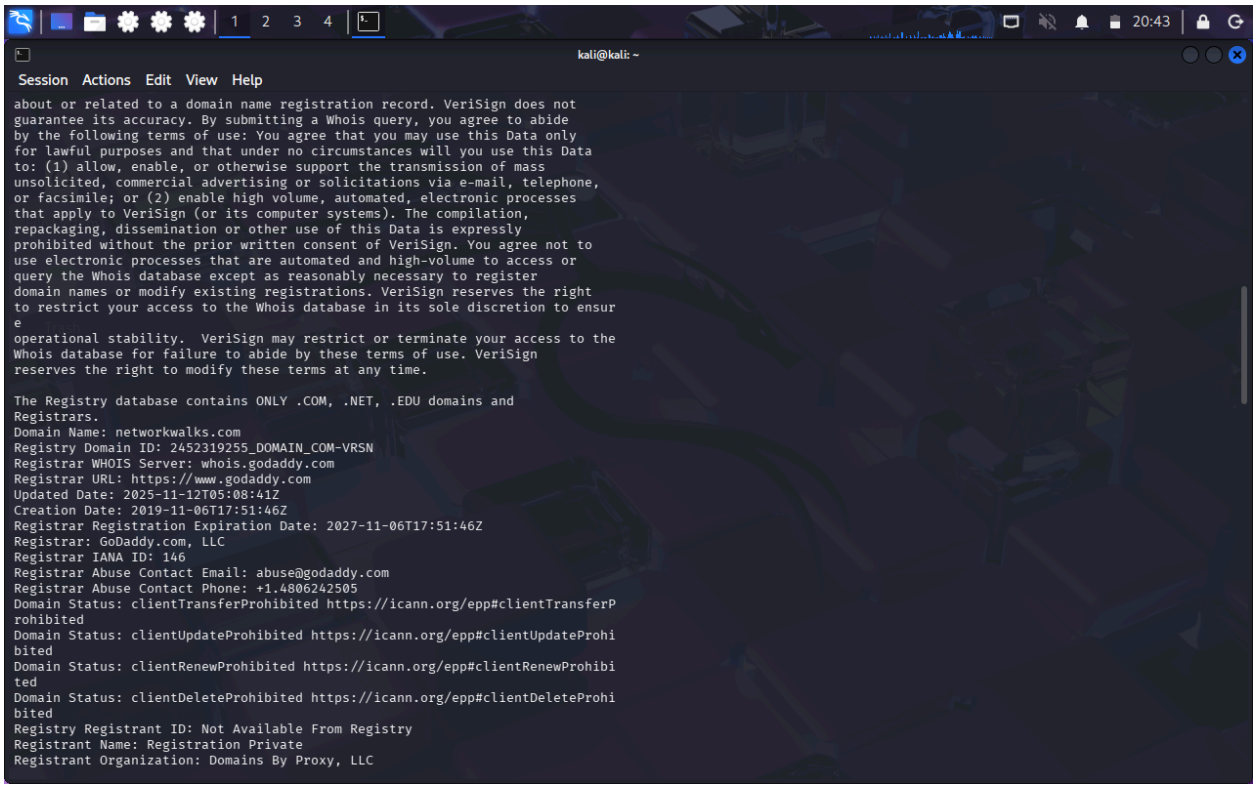
Screenshots were captured during the practical exercises to document the commands executed, tools used, and results observed.

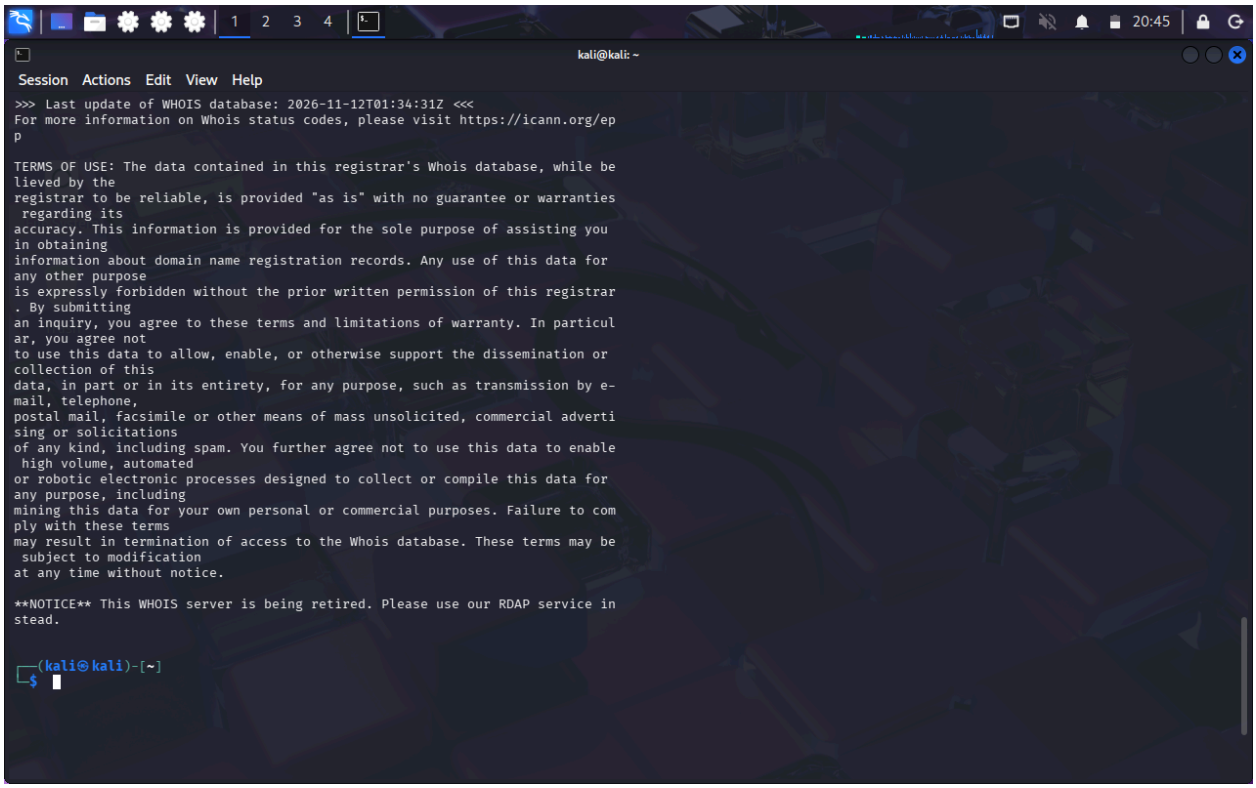
PM1 – Footprinting and Reconnaissance

- WHOIS lookup screenshots

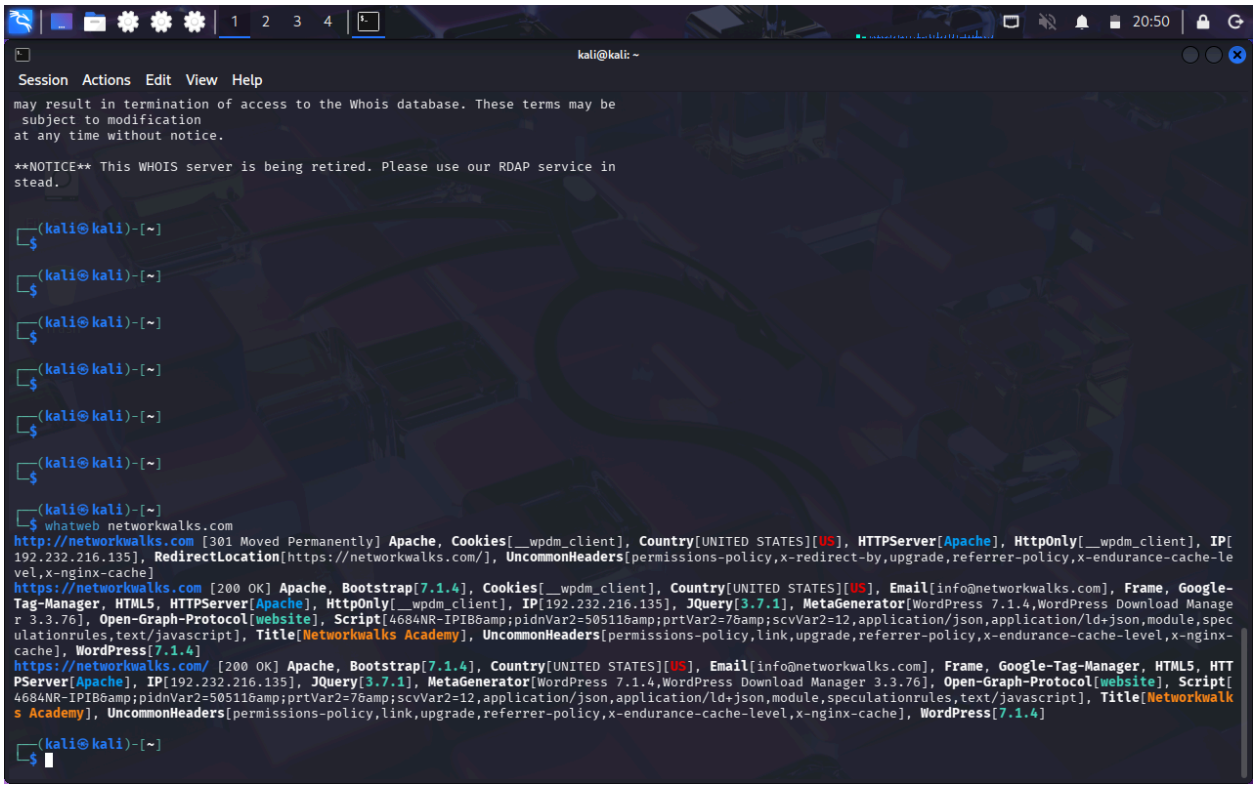



20:42
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeletePr
ohibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProh
ibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransf
erProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdatePr
ohibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wi
cf/
>>> Last update of whois database: 2026-11-12T01:34:03Z <<<
For more information on Whois status codes, please visit https://icann.org/ep
p
NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiratio
n
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry

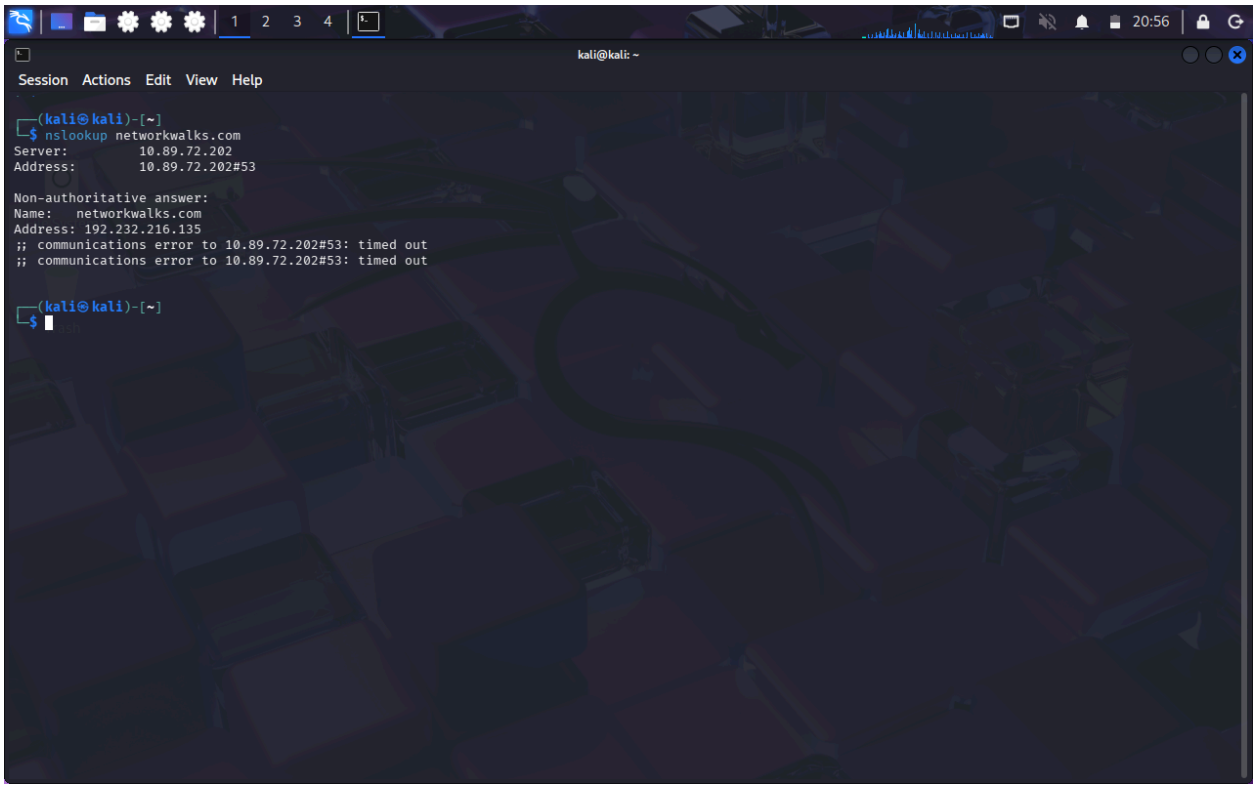

20:43
kali@kali: ~
Session Actions Edit View Help
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensur
e
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T05:08:41Z
Creation Date: 2019-11-06T17:51:46Z
Registrar Registration Expiration Date: 2027-11-06T17:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferP
rohibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhi
bited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibi
ted
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhi
bited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC



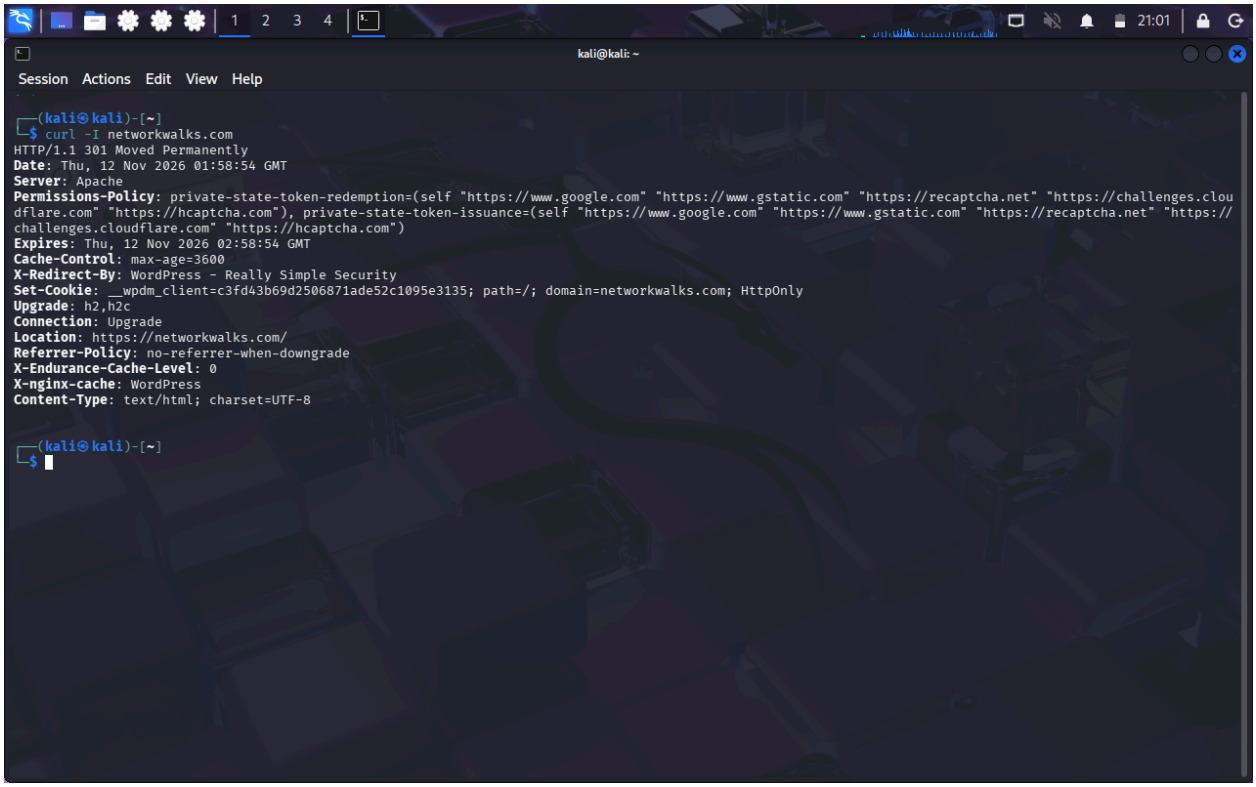
- WhatWeb scan screenshot



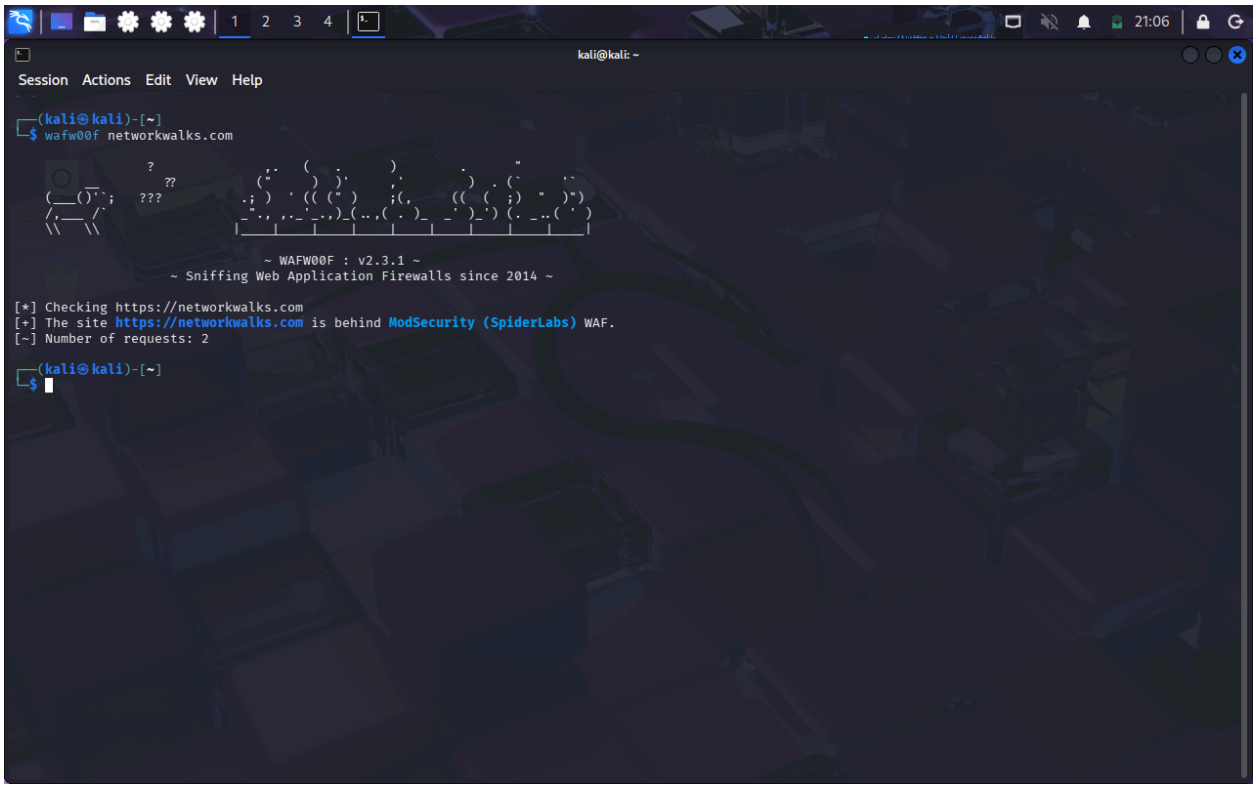
- nslookup output screenshot



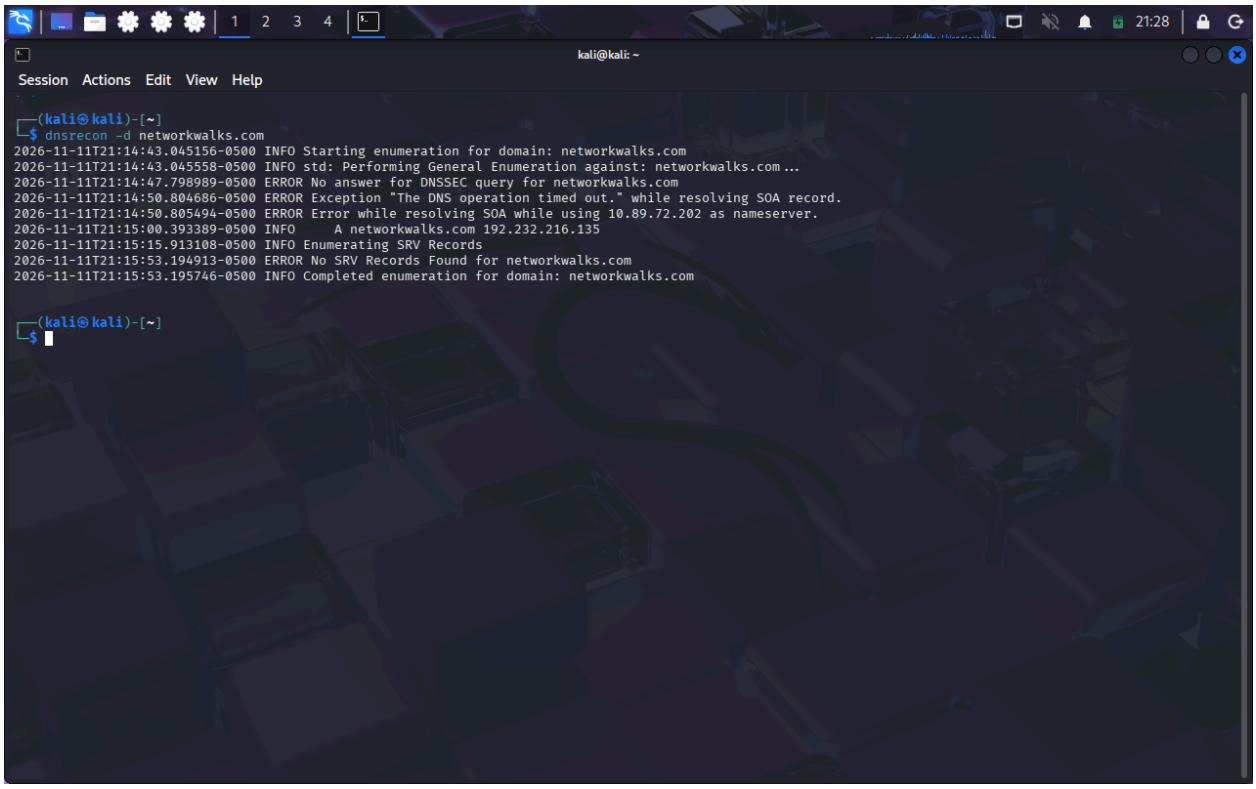
- curl HTTP response screenshot



- wafw00f scan screenshot

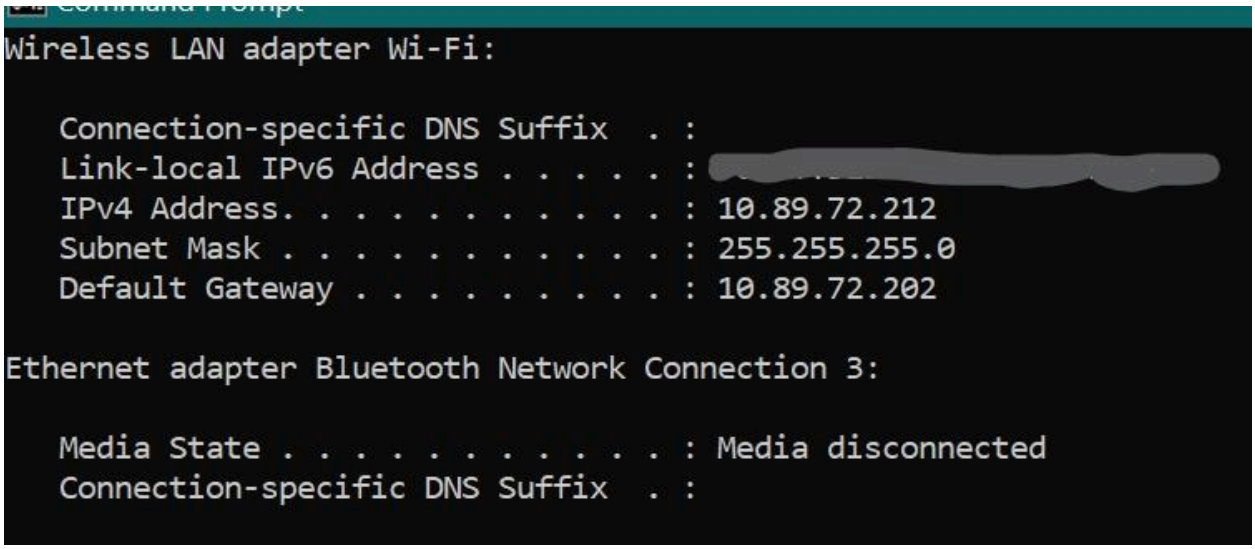


- dnsrecon output screenshot

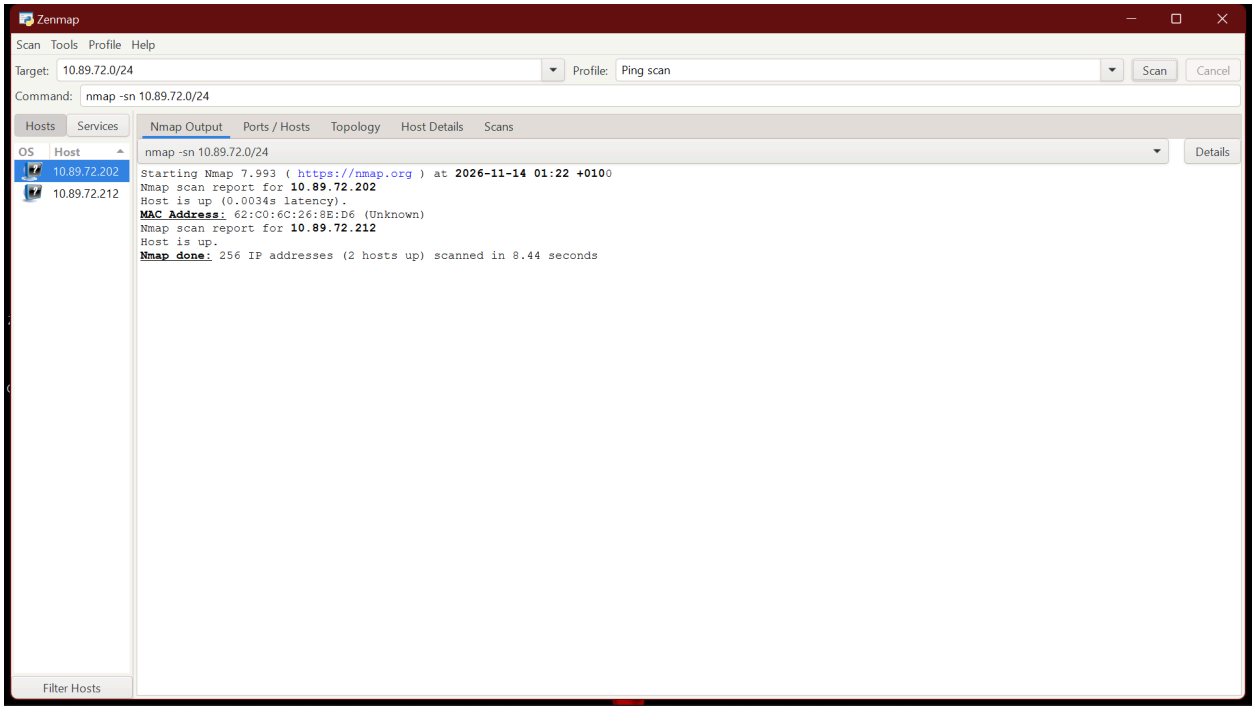


PM5 – Network Scanning and Host Discovery

- Windows `ipconfig` output



- Zenmap host discovery scan



👤 Author

Adebusola Henry
Intern B084
LinkedIn: https://www.linkedin.com/in/adebusola-henry-5a009a372/

📌 Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub